



Hello! How is it going?



too cold

sleepy not bad

пойдёт

good

bad nice

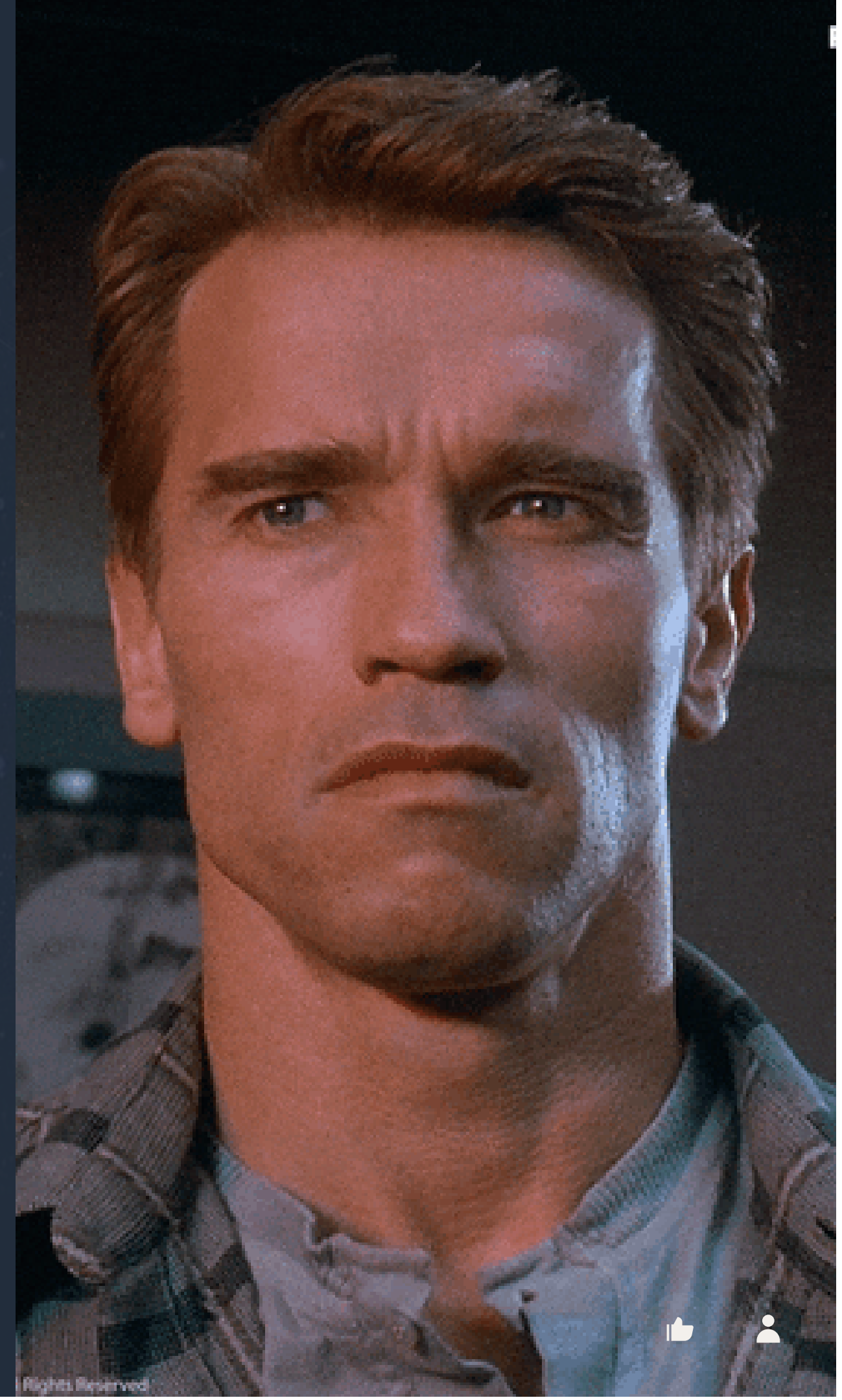
nujno bolshe zoloto

a little bit sleepy

Agenda

- Recall last session
- Discuss current topic
- Practice
- Quiz

Total Recall



What do you remember most from few previous workshops? (short answer)

jdbc and...

connneciton

SQL queries

statement

queries

What is the first step in a JDBC application?

0 ✕

Open a file stream

0 ✕

Close database connection

10 ✓

Load the JDBC driver

1 ✕

Write a query

What method sets parameters for PreparedStatement?

12 ✓



0 ✗

addParameter()

0 ✗

setRequestData()

0 ✗

setDataType()

What does PreparedStatement prevent?

12 ✓



1 ✗

Multi-threaded issues

0 ✗

Virtual machine
crashes

0 ✗

Semaphore blocking

What does the ResultSet interface represent?

0 ✗

Database connections

0 ✗

SQL commands

0 ✗

File pointers

8 ✓

Data returned by a query



Next topic

Random fact

Lemons float and limes sink in water because of differences in density and rind structure. Lemons are less dense than water due to their thick, porous rind containing air pockets. Conversely, limes have a thinner rind and a denser, more compact structure, making them denser than water and causing them to sink.



#1 Transactions Using JDBC

1.



Transaction key aspects:

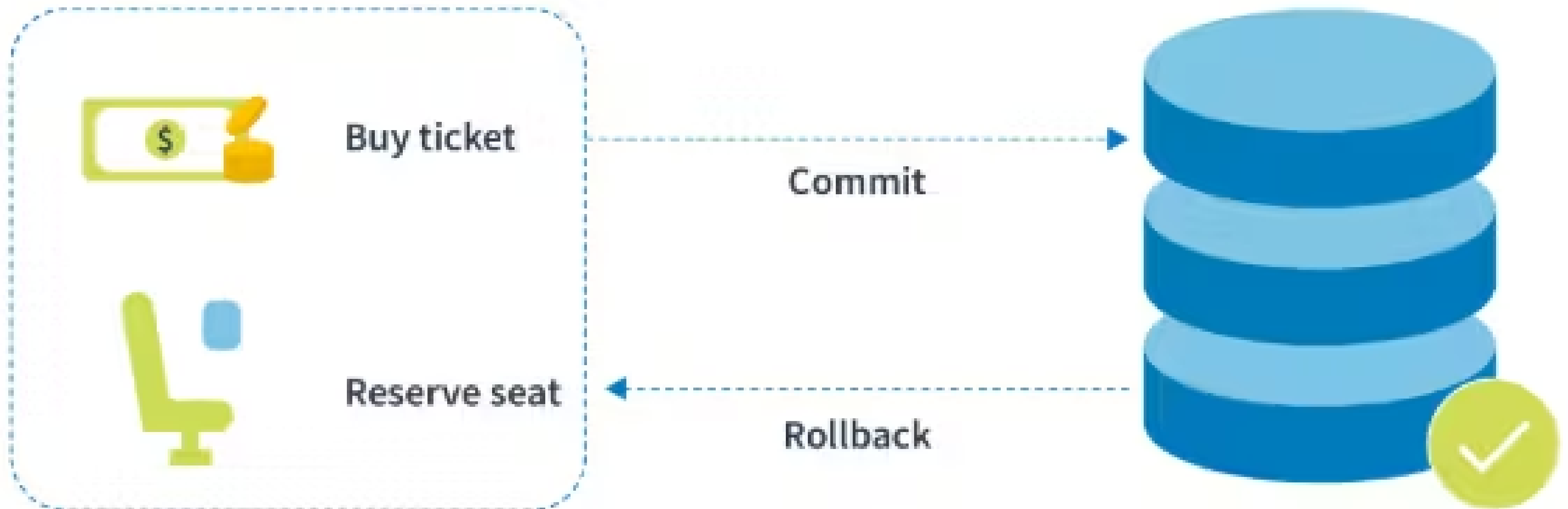
- commit - harden the state
- rollback - clear uncommitted
- auto-commit (not transactional)

ACID (principles):

- Atomicity refers to the fact that a transaction must be "all or nothing," meaning that it is either completed in its entirety or is not executed at all.
- Consistency ensures that a transaction brings a database from one stable state to another, preserving the consistency of data before and after the transaction.
- Isolation means that each transaction is independent of all other transactions. Transactions occur in isolation from one another, so one transaction does not affect the results of another.
- Durability refers to the notion that once a transaction is complete, its effects are permanent and durable. The results of a transaction must remain intact and cannot be rolled back, undone, or lost, even if the system fails or crashes.

Transaction

Database



Isolation:

- Read Uncommitted
- Read Committed
- Repeatable Read
- Serializable

Per each *reading phenomena*

[https://en.wikipedia.org/wiki/Isolation_\(database_systems\)#Read_phenomena](https://en.wikipedia.org/wiki/Isolation_(database_systems)#Read_phenomena)

<https://www.linkedin.com/pulse/read-phenomena-databases-siddharth-pandey/>

What is the role of transaction isolation levels in JDBC?

2 ✗

They specify the type of SQL queries that can be executed.

1 ✗

They control the number of database connections in a connection pool.

0 ✗

They determine the performance of JDBC connections.

11 ✓

They define how concurrent transactions are allowed to access shared data.

what

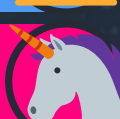


Danial Sarsenov



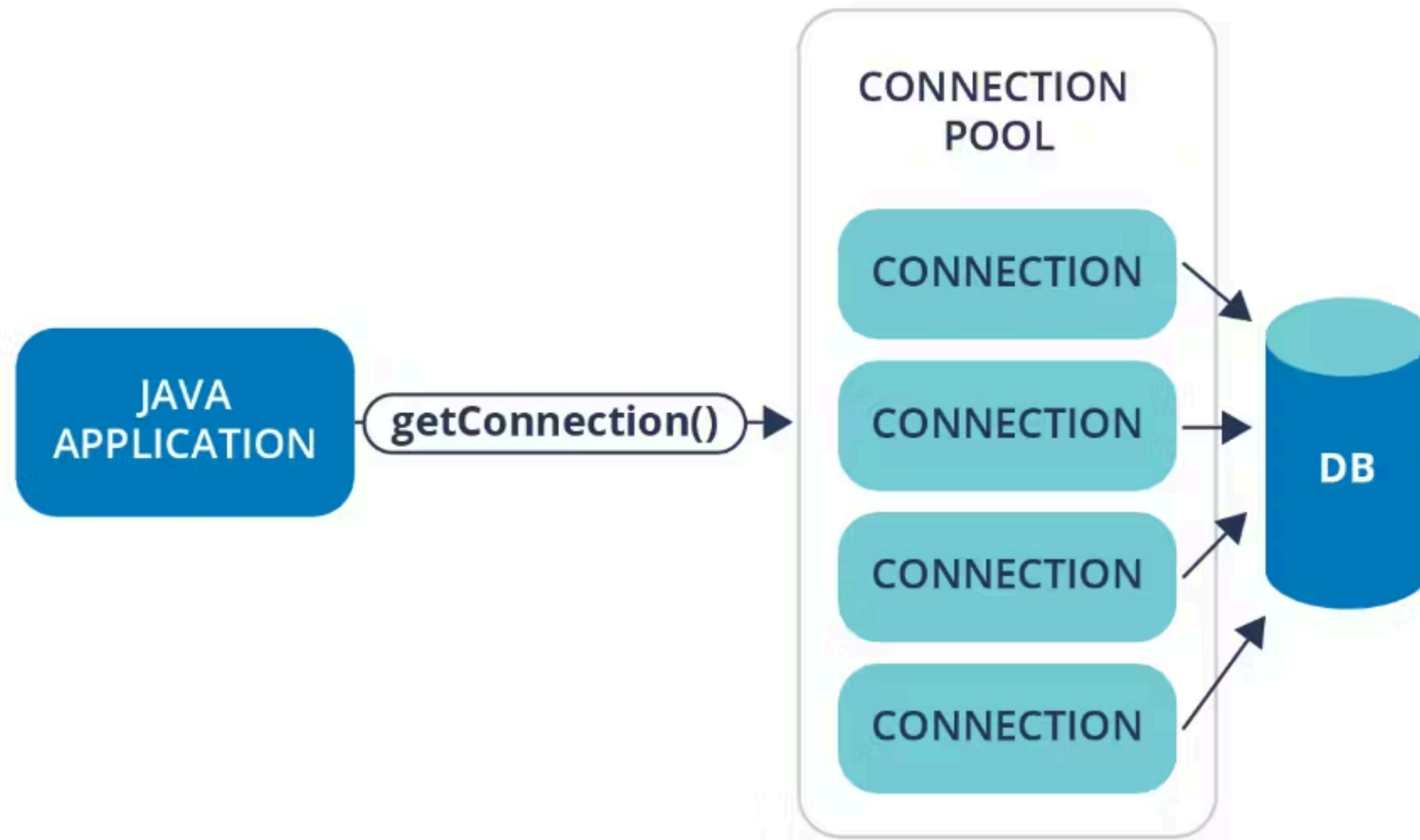
The correct answer is: The primary goal of transaction isolation is to provide a balance between data consistency and concurrency performance.

Quiz leaderboard

0 p			Kathy
0 p			Farangiz Tulkunova
0 p			Nigina Khasanova
0 p			Arslan Diveyev
0 p			Asif
0 p			Milana Ulubabyan
0 p			Vote Goat
0 p			Anvarbek Makhmudov
0 p			danial sarsenov
0 p			abdubosit

#2 Connection Pool

1. 1.



Connection pools

maintain a cache of active connections that are shared among multiple clients. When a client requires a connection, the pool checks if there is an available connection in the cache; if not, it creates a new one. Once the client is done using the connection, it returns it to the pool, making it available for reuse by other clients. This approach significantly enhances performance, scalability, and resource management for systems that access databases or remote services frequently.

#3 Metadata

1.

Database Metadata

Database metadata provides information about a database management system (DBMS) and its capabilities, including supported SQL features, database product name and version, available functions, data types, and table structures. JDBC offers the DatabaseMetaData interface to retrieve this information.

ResultSet Metadata

ResultSet metadata provides information about the structure and properties of data returned by executing SQL queries, such as column names, data types, and sizes. JDBC provides the ResultSetMetaData interface specifically for this purpose.

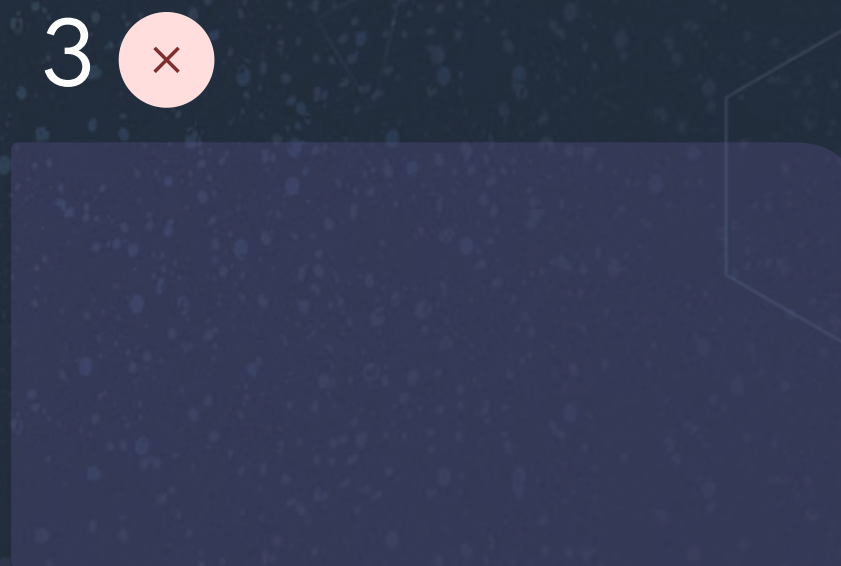
Which of the following JDBC interfaces provides methods for retrieving information about a DBMS and its capabilities?



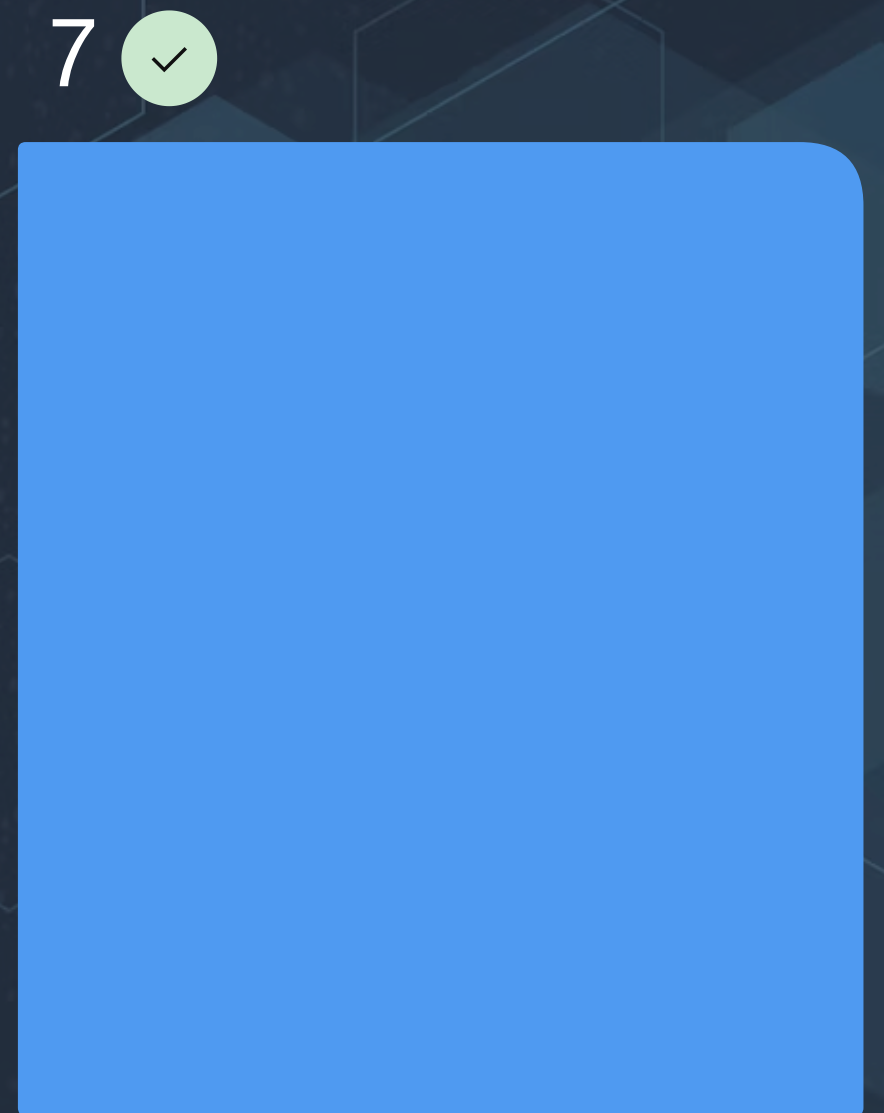
ResultSetMetaData



ConnectionMetaData



PreparedStatementMetaData



DatabaseMetaData



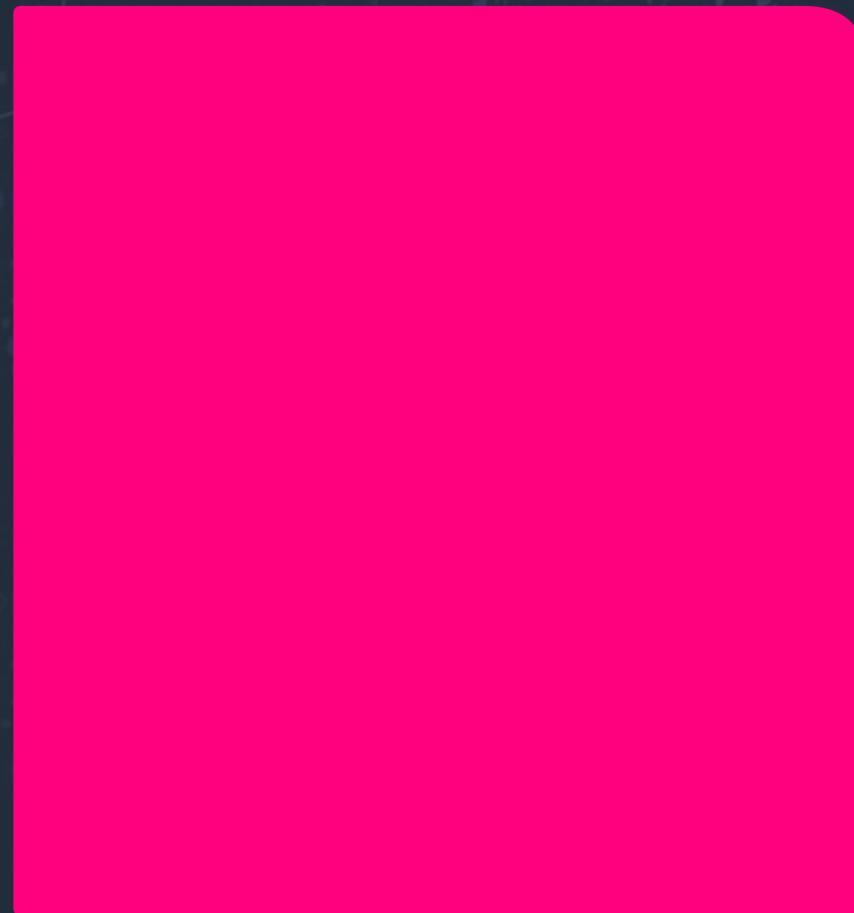
Quiz time

QUIZ



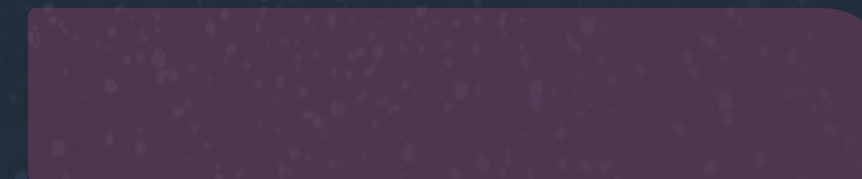
What does setAutoCommit(false) do in JDBC?

17 ✓



Disables auto-commit mode

3 ✗



Commits all transactions
immediately

0 ✗

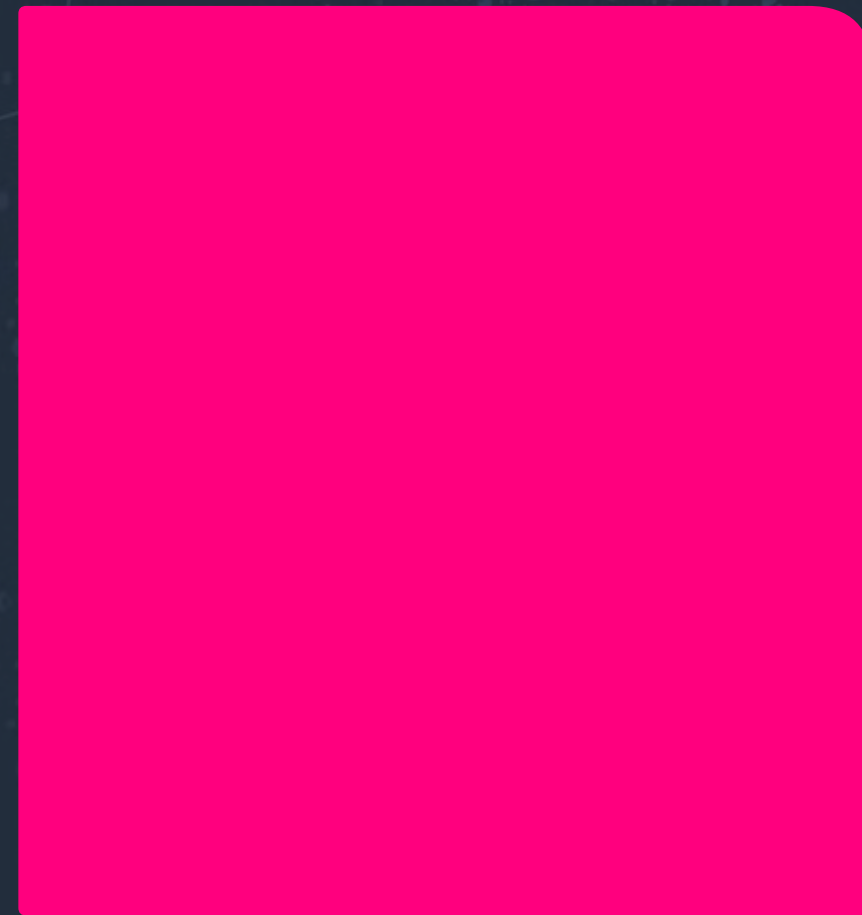
Executes the transaction

0 ✗

Rolls back current transaction

How do you commit a transaction in JDBC?

18 ✓



0 ✗

AutoCommit()

0 ✗

conn.rollback()

2 ✗

commitTransaction()

What does rollback() do in JDBC?

1 ✗

Executes all SQL statements

1 ✗

Ends JDBC connection

19 ✓

Cancels the current transaction

1 ✗

Saves the state of the database

When should transactions be used in JDBC?

17 ✓



For operations requiring atomicity

1 ✗

To test database performance

2 ✗

For selecting a single row

2 ✗

To connect to the database

How do you ensure changes are saved in a transaction?

1 ✗

Disable commit mode

0 ✗

Rollback earlier data

22 ✓

Explicitly call `commit()`

0 ✗

Use an auto-save plugin

What is a connection pool?

0 ✗

A large database table

1 ✗

Global list of queries

22 ✓

A cache of reusable database
connections

0 ✗

Encrypted database schema

What is the purpose of connection pooling?

1 ✕

To query multiple databases

0 ✕

Prevent verbose logging

21 ✓

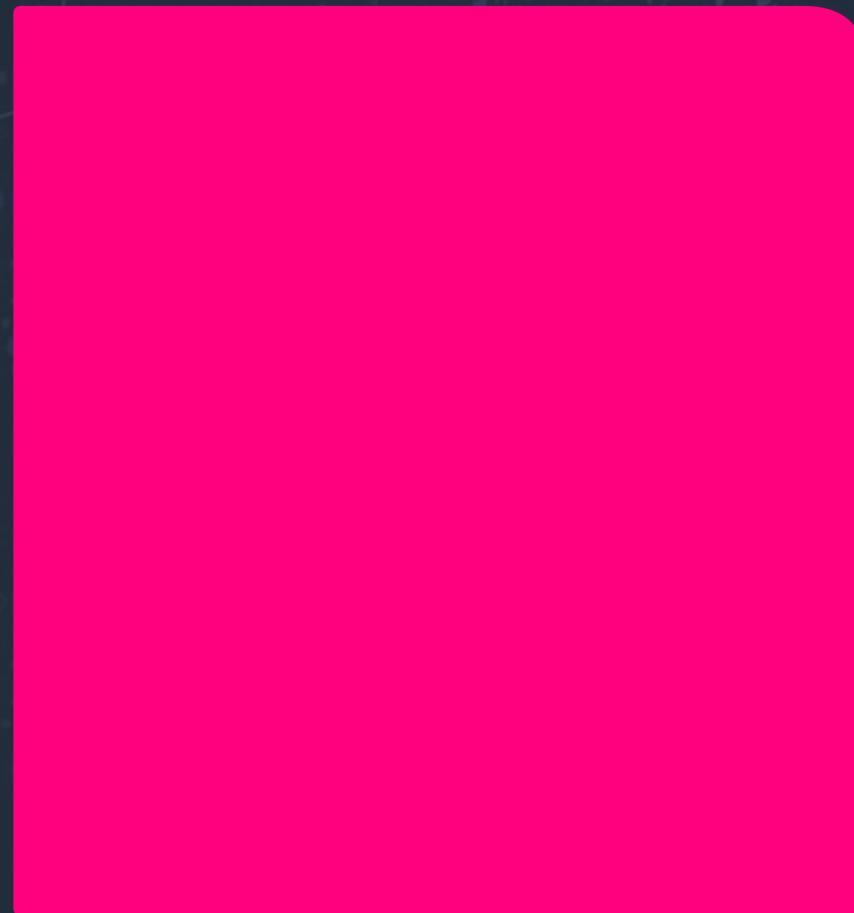
Reuse database connections
efficiently

1 ✕

Group queries by transactions

What is DatabaseMetaData?

23 ✓



2 ✗

Database schema navigator

0 ✗

A JPA configuration class

0 ✗

SQL executable command

Which example retrieves database version?

2 ✗

`conn.createStatement();`

1 ✗

`stmt.queryDatabase();`

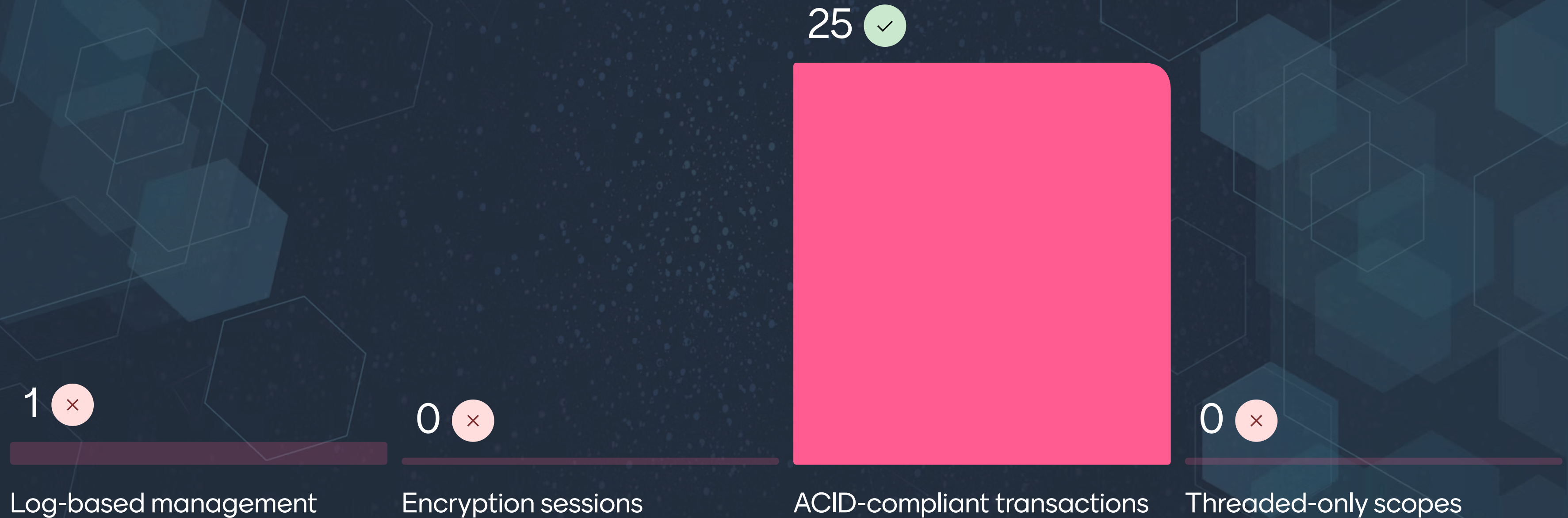
0 ✗

`conn.rollback();`

23 ✓

`metaData.getDatabaseProductVersion();`

What type of transactions does JDBC support?



Quiz leaderboard



Q&A part

1 question
0 upvotes

